Pairwise Fst

2025-03-14

```
res_dir <- "../results/s08_FST/"
```

Load and group gt

```
mad_bor_g1000_sum <-
  gt_load(
    "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_imputed.gt"
  )
mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

Impute

```
gt_set_imputed(mad_bor_g1000_sum, TRUE)
```

Replace names

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  mutate(population = case_when(
    population == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
    population == "Diego" ~ "Diego Mixed",
    TRUE ~ population
  ))
```

Pairwise Fst

```
fst_matrix <- mad_bor_g1000_sum %>%
  pairwise_pop_fst(by_locus = FALSE, method = "WC84", type = "pairwise")
```

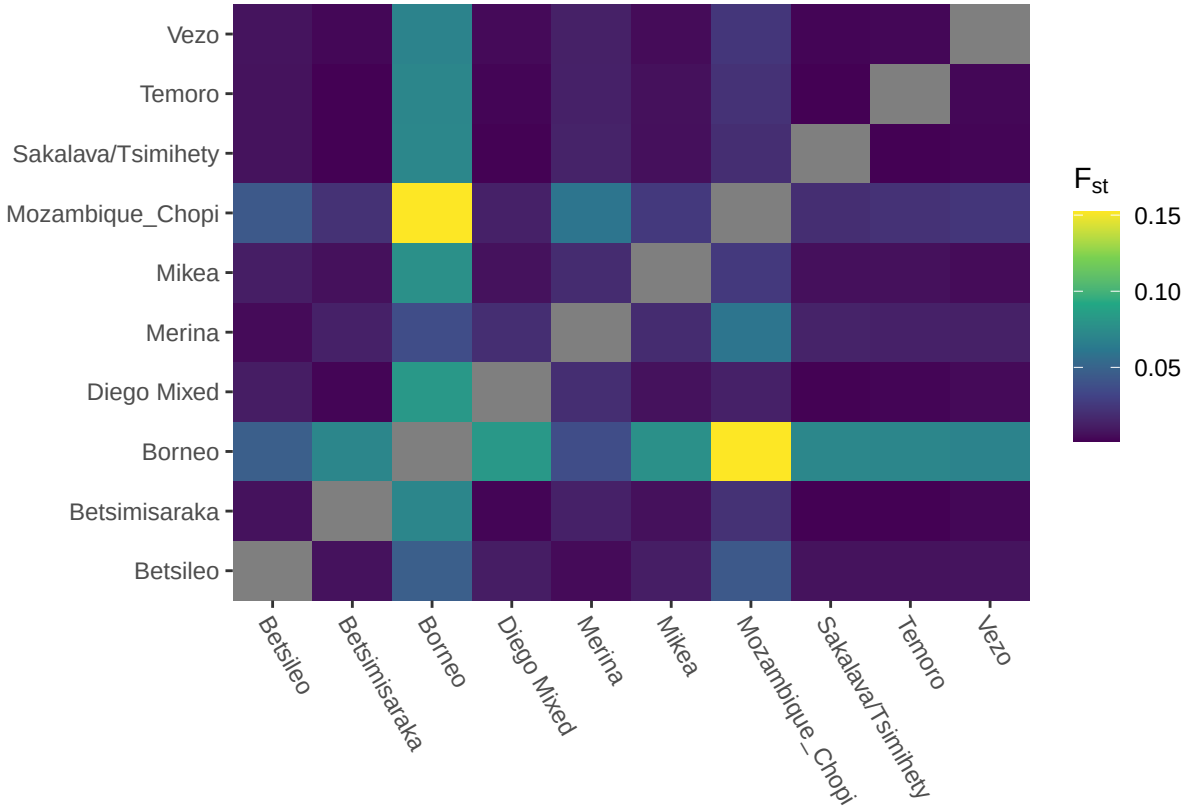
```
# subset fst matrix
sub <- c(
  "Betsimisaraka", "Merina", "Diego Mixed", "Sakalava/Tsimihety",
  "Betsileo", "Vezo", "Mikea", "Temoro", "Mozambique_Chopi", "Borneo"
)

fst_matrix <- as.data.frame(fst_matrix)
fst_matrix_sub <- as.data.frame(fst_matrix)
rm <- names(fst_matrix_sub)[(names(fst_matrix_sub) %in% sub)]
fst_matrix_sub <- fst_matrix_sub[, rm]

fst_matrix_sub <-
  subset(
    fst_matrix_sub,
    rownames(fst_matrix_sub) %in% c(
      "Betsimisaraka", "Merina", "Diego Mixed",
      "Sakalava/Tsimihety", "Betsileo",
      "Vezo", "Mikea", "Temoro",
      "Mozambique_Chopi", "Borneo"
    )
  )
```

```
ggheatmap(fst_matrix_sub) +
  scale_fill_viridis_c(expression("F"[st])) +
  theme(axis.text.x = element_text(angle = -60, hjust = 0))
```

```
## > heatmap built with `geom_tile()`
```



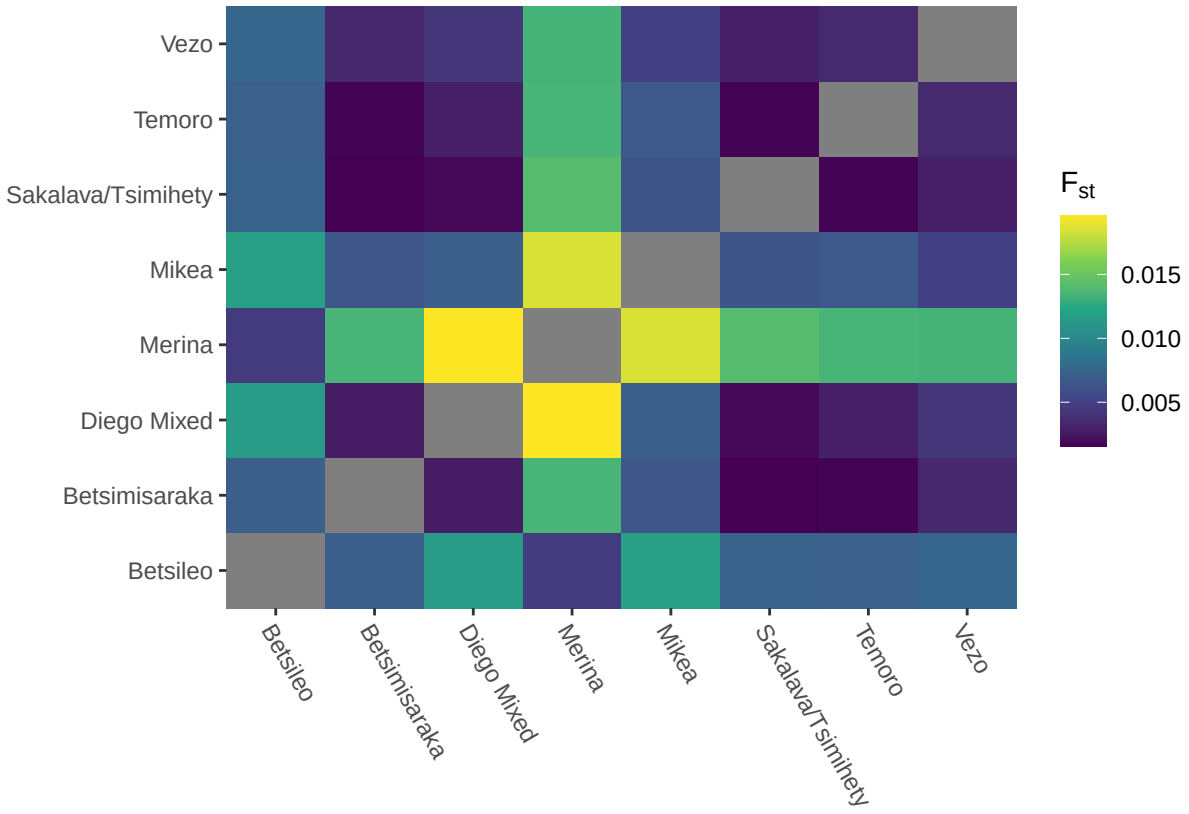

```
sub <- c(
  "Betsimisaraka", "Merina", "Diego Mixed", "Sakalava/Tsimihety",
  "Betsileo", "Vezo", "Mikea", "Temoro"
)

fst_matrix_sub <- as.data.frame(fst_matrix)
rm <- names(fst_matrix_sub)[(names(fst_matrix_sub) %in% sub)]
fst_matrix_sub <- fst_matrix_sub[, rm]


fst_matrix_sub <-
  subset(
    fst_matrix_sub,
    rownames(fst_matrix_sub) %in% c(
      "Betsimisaraka", "Merina", "Diego Mixed",
      "Sakalava/Tsimihety", "Betsileo",
      "Vezo", "Mikea", "Temoro"
    )
  )

fst_merina_together <- ggheatmap(fst_matrix_sub) +
  scale_fill_viridis_c(expression("F"[st])) +
  theme(axis.text.x = element_text(angle = -60, hjust = 0))
fst_merina_together
```

```
## > heatmap built with `geom_tile()`
```




```
ggsave(paste0(res_dir, "fst_merina_together.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
print(fst_matrix_sub)
```

```
##                      Betsileo Betsimisaraka Diego Mixed      Merina
## Betsileo                   NA   0.006938240 0.011408492 0.004543445
## Betsimisaraka      0.006938240            NA 0.002708546 0.013443895
## Diego Mixed        0.011408492   0.002708546          NA 0.019549327
## Merina             0.004543445   0.013443895 0.019549327          NA
## Mikea              0.011741217   0.006362411 0.006938704 0.018494238
## Sakalava/Tsimihety 0.007204705   0.001584122 0.001863413 0.014011642
## Temoro             0.007057056   0.001684712 0.002960115 0.013440671
## Vezo               0.007493045   0.003357342 0.004372498 0.013305260
##                           Mikea Sakalava/Tsimihety      Temoro        Vezo
## Betsileo           0.011741217        0.007204705 0.007057056 0.007493045
## Betsimisaraka      0.006362411        0.001584122 0.001684712 0.003357342
## Diego Mixed        0.006938704        0.001863413 0.002960115 0.004372498
## Merina             0.018494238        0.014011642 0.013440671 0.013305260
## Mikea                       NA        0.006247245 0.006502478 0.004855969
## Sakalava/Tsimihety 0.006247245                 NA 0.001694304 0.002960526
## Temoro             0.006502478        0.001694304          NA 0.003485504
## Vezo               0.004855969        0.002960526 0.003485504          NA
```

Population level FST for Merina subsets

```
qmats <- read_q_files("../data/intermediate/admixture")

fam <- read.table(
  "../data/intermediate/admixture/admixture_supervised_subset.fam",
  header = FALSE
)

mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")

qmats$id <- fam$V2

# subset mad_bor_g1000_sum to only individuals in qmats$id
mad_bor_g1000_sum_sub <- mad_bor_g1000_sum %>%
  filter(id %in% qmats$id)

# tidy with the gt
mad_bor_g1000_sum_sub <- augment(get_q_matrix(qmats, k = 2, run = 1),
  data = mad_bor_g1000_sum_sub
)
# filter to merina
merina <- mad_bor_g1000_sum_sub %>% filter(population == "Merina")

# find which .Q column represents Asian ancestry
asian_ancestry_q <- mad_bor_g1000_sum_sub %>%
  filter(population == "Borneo") %>%
  # Get median value of Q1 and Q2
  summarise(.Q1 = median(.Q1), .Q2 = median(.Q2)) %>%
  # Pivot in order to subset
  pivot_longer(
    cols = c(".Q1", ".Q2"),
    names_to = "column_name",
    values_to = "median_val"
  ) %>%
  # Find the row with the max value
  slice_max(median_val, n = 1) %>%
  # get the column name
  pull(column_name)


# split groups
merina$population <- ifelse(merina %>% select(all_of(asian_ancestry_q)) > 0.5,
  "Merina A",
  "Merina B"
)
merina <- merina[, c("id", "population")]

# match V2 and merina$id and replace the population column in merina_fam with
# the population column in merina
matching_m <- match(mad_bor_g1000_sum$id, merina$id)
mad_bor_g1000_sum$population <- ifelse(!is.na(matching_m),
```

```
  merina$population[matching_m], mad_bor_g1000_sum$population
)

# count number of MerinaA
cat(
  "Merina group > 0.5",
  length(which(mad_bor_g1000_sum$population == "Merina A")),
  "\n"
)
```

```
## Merina group > 0.5 23
```

```
cat(
  "Merina group < 0.5",
  length(which(mad_bor_g1000_sum$population == "Merina B")),
  "\n"
)
```

```
## Merina group < 0.5 11
```

Pairwise Fst

```
mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  mutate(population = case_when(
    population == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
    population == "Diego" ~ "Diego Mixed",
    TRUE ~ population
  ))

mad_bor_g1000_sum <- mad_bor_g1000_sum %>%
  filter(population %in% c(
    "Betsimisaraka", "Merina A", "Merina B", "Diego Mixed",
    "Sakalava/Tsimihety", "Betsileo", "Vezo",
    "Mikea", "Temoro"
  ))

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% group_by(population)
```

```
fst_matrix <- mad_bor_g1000_sum %>%
  pairwise_pop_fst(by_locus = FALSE, method = "WC84", type = "pairwise")

write.csv(fst_matrix, paste0(res_dir, "fst_matrix_full.txt"),
  row.names = TRUE
)
```

Pairwise Fst heatmap with dendrogram and Merina split

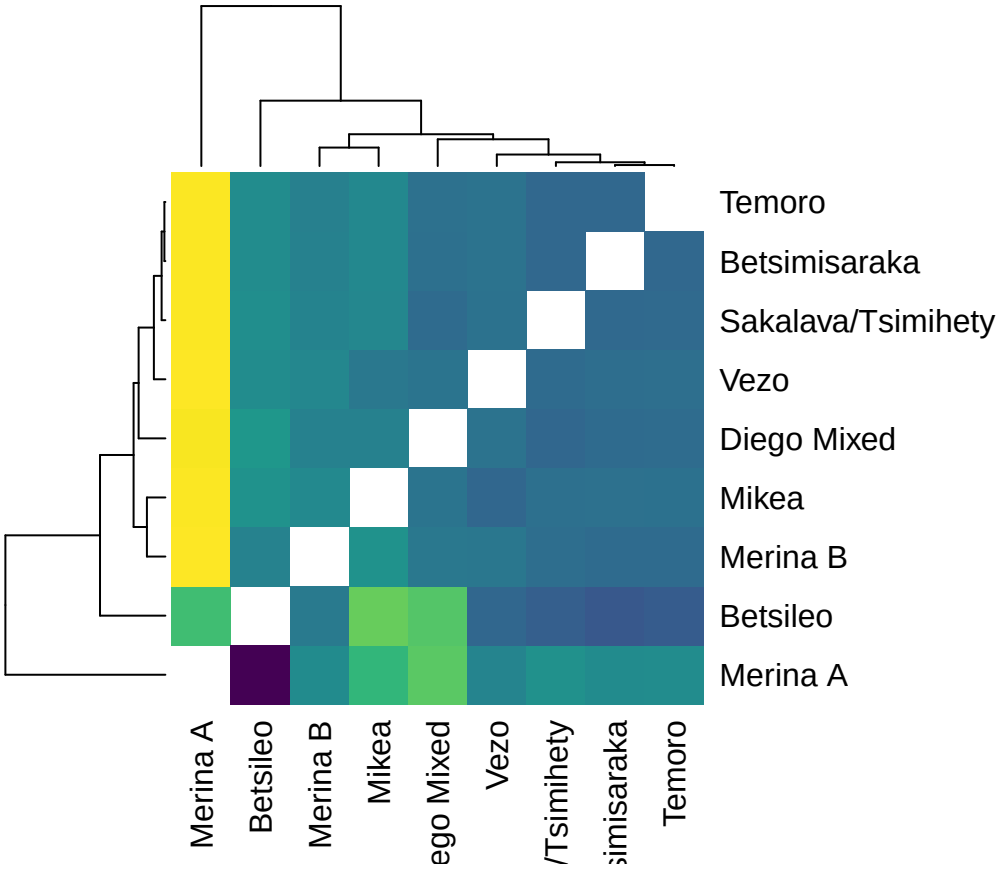
https://bioinformatics.ccr.cancer.gov/docs/data-visualization-with-r/Lesson5_intro_to_ggplot/

```
pheatmap(fst_matrix,
  color = viridis(256),
  angle_col = 45,
  filename = paste0(res_dir, "fst_dendrogram_merina_split.jpg"),
  width = 7, height = 5, dpi = 300
)
```

Other heatmap versions

https://stackoverflow.com/questions/38727324/adding-a-dendrogram-to-a-ggplot2-heatmap https://yunuuuu.github.io/ggalign-book/complexheatmap-single-heatmap.html

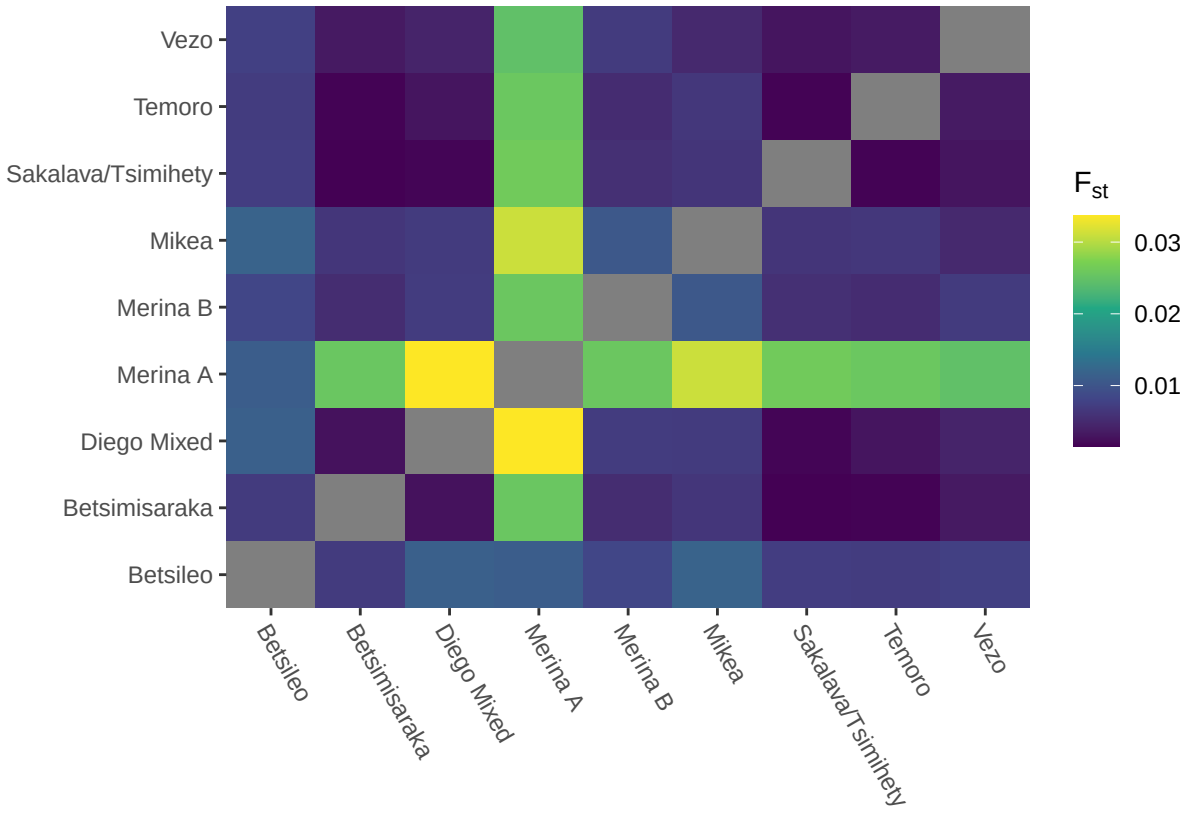
```
heatmap(fst_matrix, col = viridis(256)) +
  theme(axis.text.x = element_text(angle = -60, hjust = 0))
```




```
## NULL
```

```
ggheatmap(fst_matrix) +
  scale_fill_viridis_c(name = expression("F"[st])) +
  theme(axis.text.x = element_text(angle = -60, hjust = 0))
```

```
## > heatmap built with `geom_tile()`
```

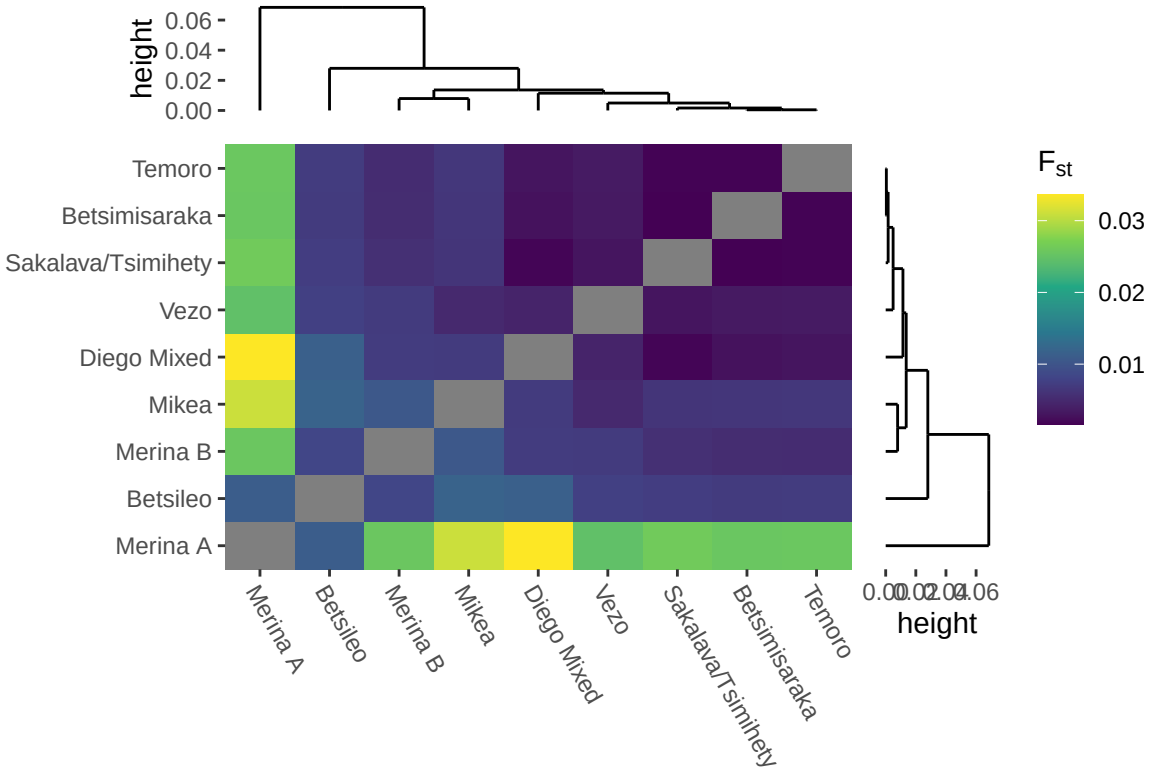



```
ggheatmap(fst_matrix) +
  scale_fill_viridis_c(name = expression("F"[st])) +
  anno_right(size = unit(15, "mm")) +
  align_dendro(reorder_dendrogram = TRUE) +
  theme(axis.text.y = element_blank()) +

  anno_top(size = unit(15, "mm")) +
  align_dendro(reorder_dendrogram = TRUE) +
  theme(axis.text.x = element_blank()) +

  quad_active() +
  theme(
    plot.margin = margin(),
    axis.text.x = element_text(angle = -60, hjust = 0)
  )
```

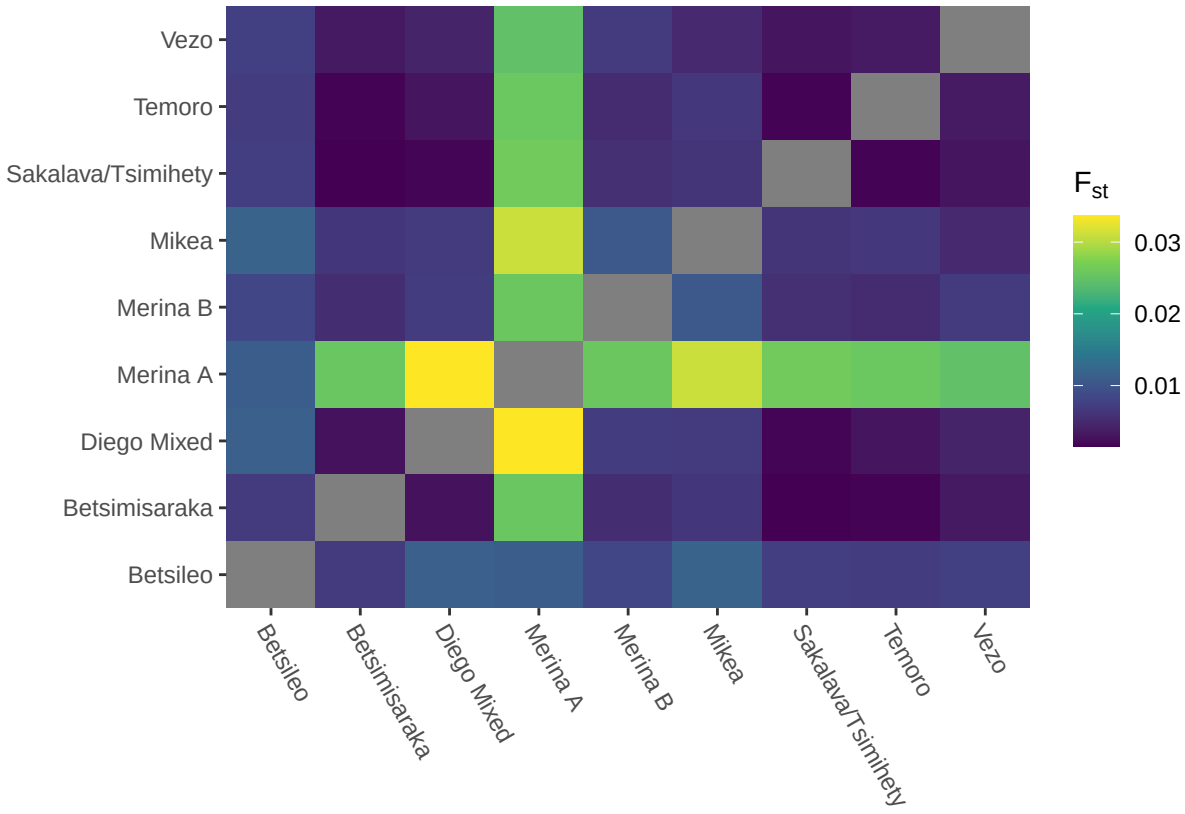
```
## > heatmap built with `geom_tile()`
```



Merina split with no dendrogram

```
fst_merina_split <- ggheatmap(fst_matrix) +
  scale_fill_viridis_c(name = expression("F"[st])) +
  theme(axis.text.x = element_text(angle = -60, hjust = 0))
fst_merina_split
```

```
## > heatmap built with `geom_tile()`
```




```
ggsave(paste0(res_dir, "fst_merina_split.jpg"),
  width = 7, height = 5, dpi = 300
)
```

Compare Fst to difference in African ancestry proportion between populations

```
f4_ratios <- read.table("../results/s06_b_f_stats/f4_ratio_african_admixture",
  header = TRUE, sep = "\t"
)
f4_ratios <- f4_ratios %>%
  mutate(pop3 = case_when(
    pop3 == "MerinaA" ~ "Merina A",
    pop3 == "MerinaB" ~ "Merina B",
    pop3 == "Sakalava_Tsimihety" ~ "Sakalava/Tsimihety",
    pop3 == "Diego" ~ "Diego Mixed",
    TRUE ~ pop3
  ))

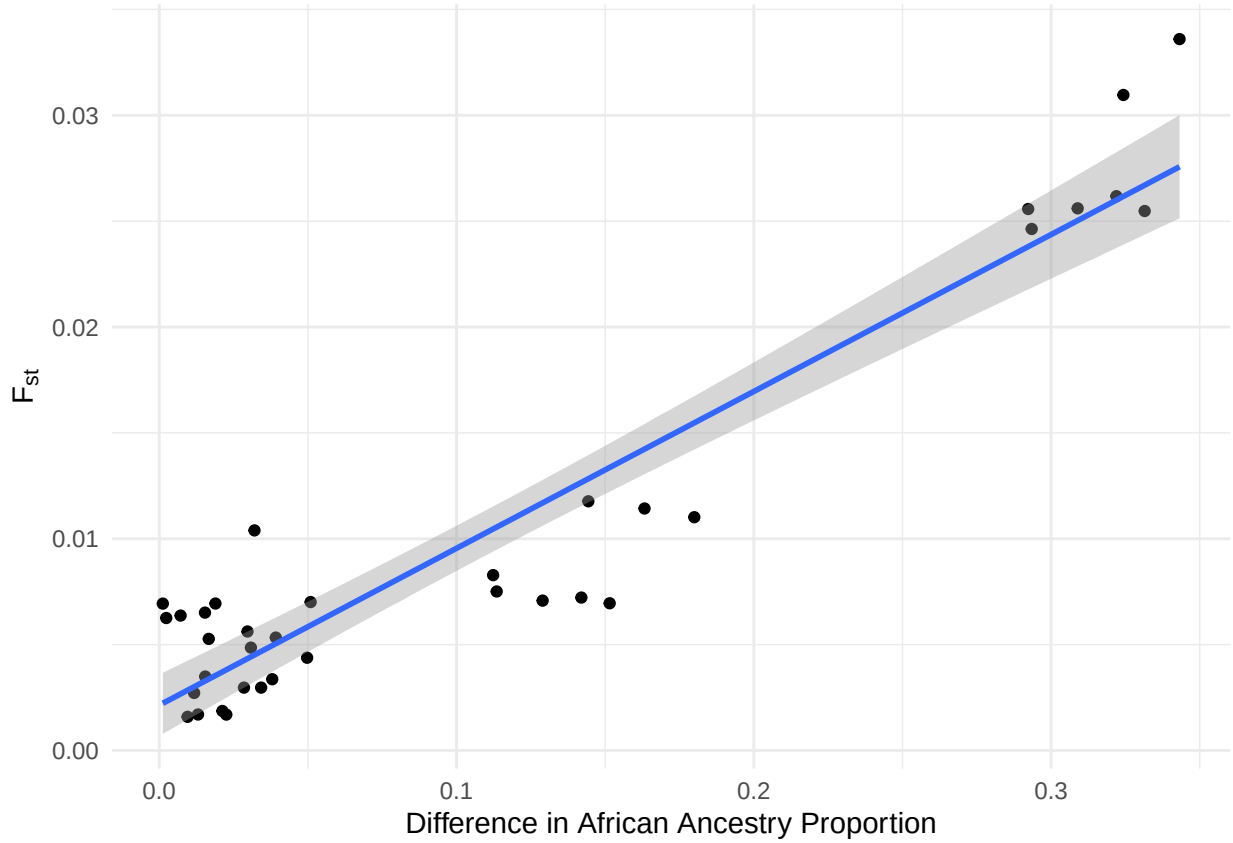

fst_matrix_tidy <- mad_bor_g1000_sum %>%
  pairwise_pop_fst(by_locus = FALSE, method = "WC84", type = "tidy")
```

```r
fst_matrix_tidy <- fst_matrix_tidy %>%
  mutate(
    anc_pop_1 = f4_ratios[match(fst_matrix_tidy$population_1, f4_ratios$pop3), "alpha"],
    anc_pop_2 = f4_ratios[match(fst_matrix_tidy$population_2, f4_ratios$pop3), "alpha"]
  )

fst_matrix_tidy <- fst_matrix_tidy %>%
  mutate(
    diff = abs(anc_pop_1 - anc_pop_2),
    pops = paste(population_1, population_2, sep = "_")
  )

ggplot(fst_matrix_tidy, aes(x = diff, y = value)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(
    x = "Difference in African Ancestry Proportion",
    y = expression("F"[st])
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```r
ggsave(paste0(res_dir, "fst_vs_ancestry_diff.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```r
# get proportion of Fst variance explained by ancestry
model <- lm(value ~ diff, data = fst_matrix_tidy)
summary(model)
```

```
##
## Call:
## lm(formula = value ~ diff, data = fst_matrix_tidy)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -0.0064127 -0.0017445 -0.0000539  0.0018070  0.0060304
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.0021387  0.0007144   2.994  0.00511 **
## diff        0.0740942  0.0045587  16.253  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.00314 on 34 degrees of freedom
## Multiple R-squared:  0.886,  Adjusted R-squared:  0.8826
## F-statistic: 264.2 on 1 and 34 DF,  p-value: < 2.2e-16
```

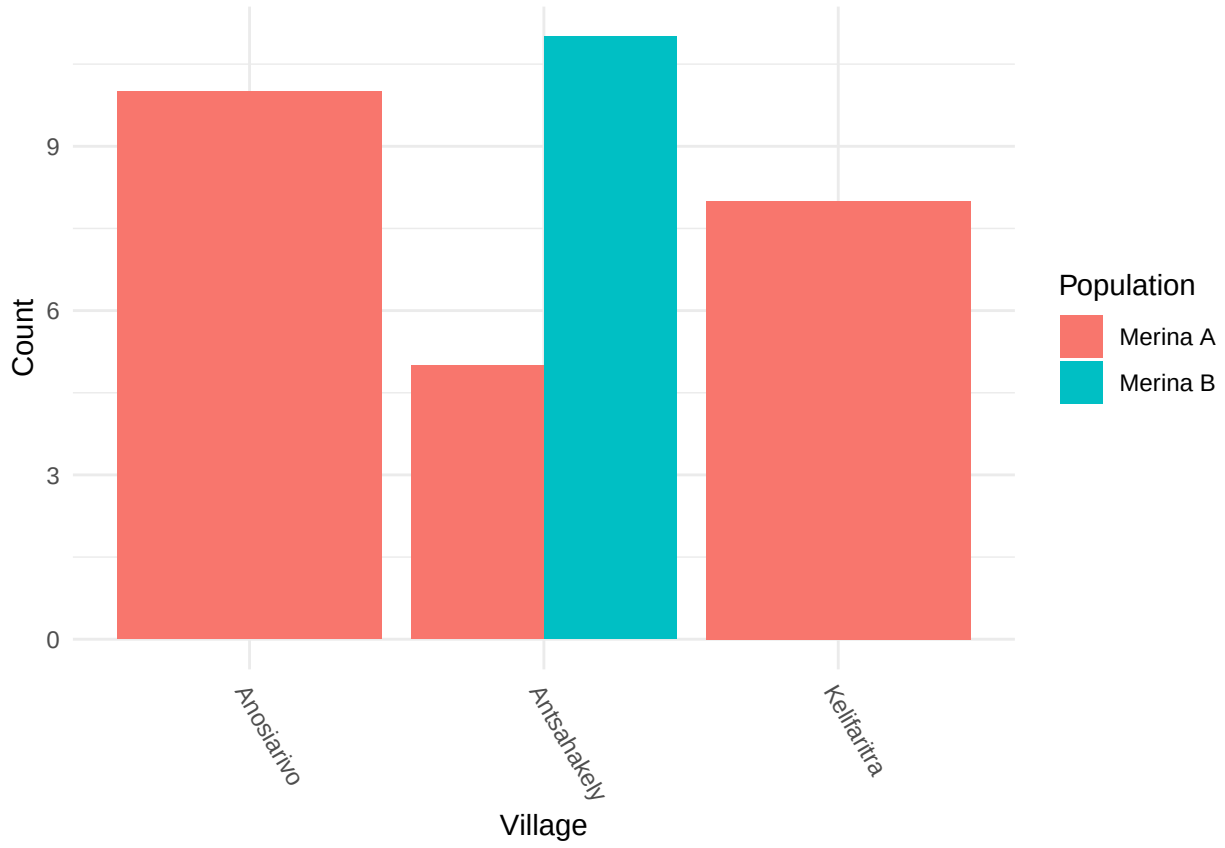
Check spatial distribution of Merina A and Merina B

```r
samples <-
  read.csv("../data/raw/data_input/Malagasy_pop_gen_sample_sheet_2015_cleaned.csv")
samples <- samples %>%
  mutate(Sample.ID = paste0(Sample.ID, "RR"))

mad_bor_g1000_sum_samples <- mad_bor_g1000_sum %>%
  left_join(samples, by = c("id" = "Sample.ID"))

spatial_dist <- mad_bor_g1000_sum_samples %>%
  filter(population %in% c("Merina A", "Merina B")) %>%
  select(Village, population)

ggplot(spatial_dist, aes(x = Village, fill = population)) +
  geom_bar(position = "dodge") +
  labs(x = "Village", y = "Count", fill = "Population") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = -60, hjust = 0))
```


Count
9
6
3
0
Anosiarivo
Antsahakely
Kelifaritra
Village
Population
Merina A
Merina B